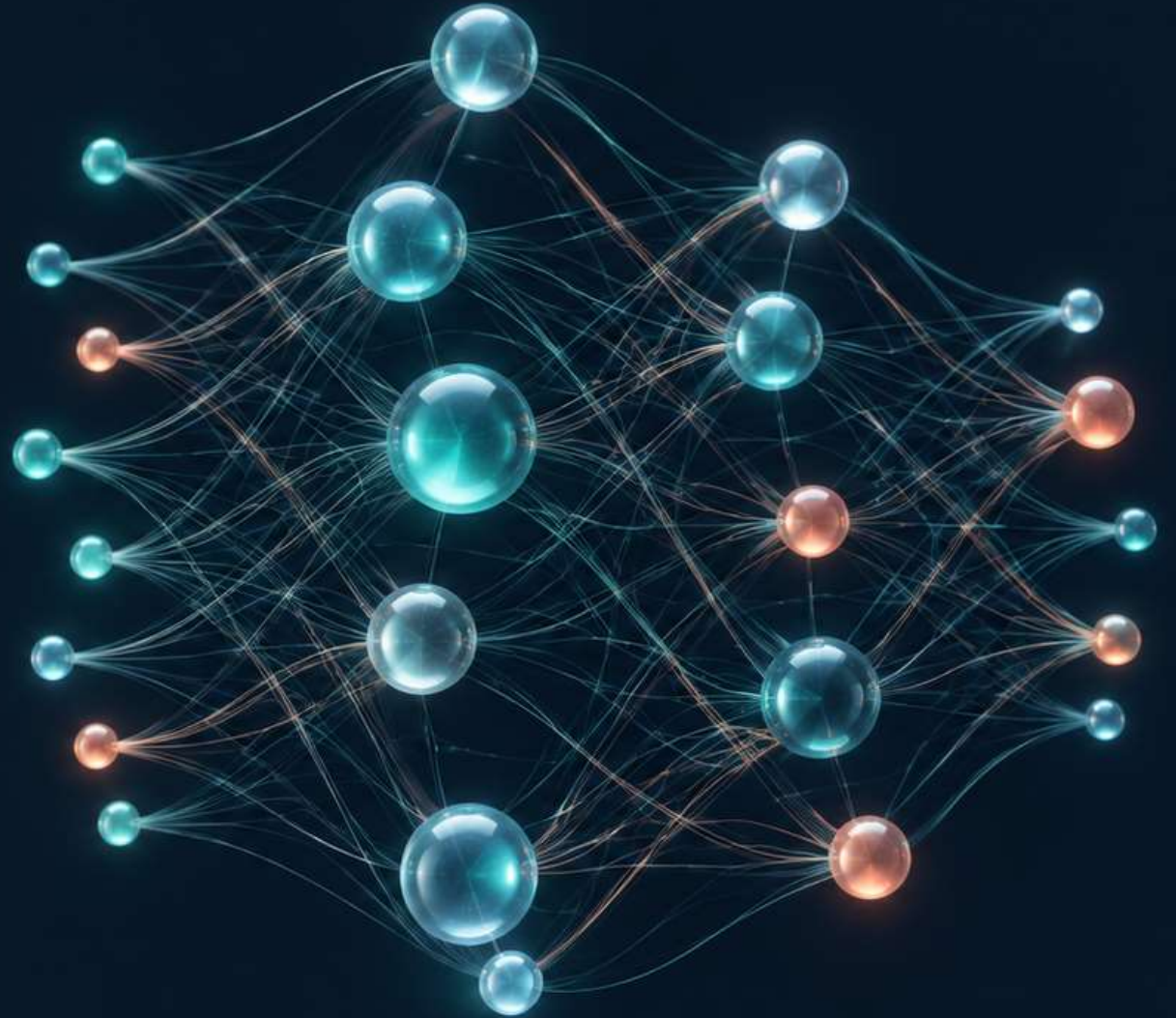


Neural networks

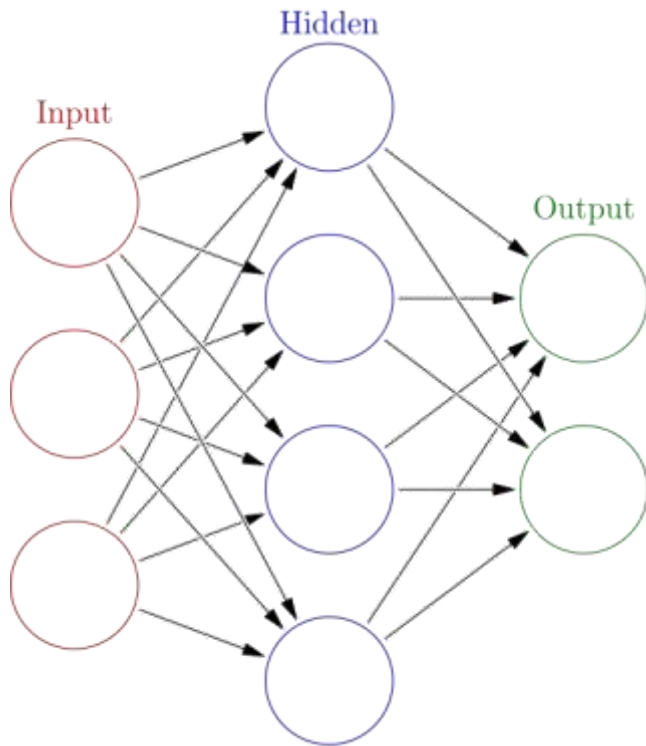
How models learn from examples

A visual introduction



Inside a neural network

A neural network learns a mapping between inputs and outputs.



Input layer

Receives numbers, such as pixel values.

Hidden layers

Combine signals into learned representations.

Output layer

Produces a prediction, such as a class score.

What one neuron computes

1 Combine the inputs

$$z = w_1x_1 + w_2x_2 + b$$

Inputs x carry data. Weights w and bias b are learned.

2 Apply an activation function

$$a = \text{ReLU}(z) = \max(0, z)$$

Nonlinearity lets layers learn more complex relationships.

How training works

Training repeatedly adjusts weights and biases to reduce a loss.

01

Predict

Run a batch of examples through the network.

02

Measure loss

Compare predictions with target answers.

03

Backpropagate

Calculate how each parameter affects loss.

04

Update

Use an optimizer to adjust the parameters.

Repeat over many batches. The learning rate controls the update size.

Example: recognizing a handwritten digit

A digit classifier assigns a score to each possible answer, 0 through 9.

28 × 28

pixel image

784 pixel values become
the model's input.

10

possible digit classes

Softmax converts output scores
into a probability distribution.

Prediction = the class with the largest probability

Common architectures and applications

Different structures suit different kinds of data.

Architecture	Main idea	Example use
Dense network (MLP)	Connect every neuron to the next layer	Tabular prediction
Convolutional network	Reuse filters across local regions	Image recognition
Transformer	Use attention to combine information across tokens	Translation and language generation

The goal is performance on new data

Overfitting happens when training improves but generalization gets worse.



Illustrative curves, not experimental results

Train

Learn parameters on training data.

Validate

Tune choices and watch for overfitting.

Test

Evaluate on held-out examples.